

ASSIGNMENT 01

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Section: 07

1.

Given that, $f(x) = 1 - x^2$, $[-3, -1]$

Now, $\Delta x = \frac{b-a}{n}$

$$= \frac{-1+3}{n} = \frac{2}{n}$$

Let, x_k^* is the right endpoint

$$x_k^* = a + k \Delta x$$

$$= -3 + k \cdot \frac{2}{n}$$

$$= -3 + \frac{2k}{n}$$

$$f(x_k^*) = 1 - \left(-3 + \frac{2k}{n}\right)^2$$

$$= 1 - \left\{ (-3)^2 + 3(-3) \frac{2k}{n} + \left(\frac{2k}{n}\right)^2 \right\}$$

$$= 1 + 27 - \frac{54k}{n} + \frac{36k^2}{n^2} - \frac{8k^3}{n^3}$$

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(28 - \frac{54k}{n} + \frac{36k^2}{n^2} - \frac{8k^3}{n^3} \right) \frac{2}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left\{ \frac{1}{n} (56) - \frac{1}{n^2} k (108) + \frac{1}{n^3} k^2 (72) - \frac{1}{n^4} k^3 (16) \right\}$$

$$= 56 - \frac{1}{2} (108) + \frac{1}{3} (72) - \frac{1}{4} (16)$$

$$= 22$$

(Ans)

Q Given, $\int_{\pi/6}^{3\pi/2} \cot^5\left(\frac{x}{6}\right) \sec^2\left(\frac{x}{6}\right) dx$

Let $= \int_{\pi/6}^{3\pi/2} \frac{1}{\tan^5 \frac{x}{6}} \cdot \sec^2 \frac{x}{6} dx$

Let, $\tan \frac{x}{6} = u$

$\Rightarrow \sec^2 \frac{x}{6} dx = 6 du$

$\int \frac{1}{u^5} 6 du = 6 \int \frac{du}{u^5}$

$= \frac{6}{-4} \left[\frac{1}{u^4} \right] = -\frac{3}{2} \frac{1}{u^4} + C$

Now, $\int_{\pi/6}^{3\pi/2}$

$\frac{1}{\tan^5 \frac{x}{6}} \sec^2 \frac{x}{6} dx = \left[-\frac{3}{2} \frac{1}{\tan^4 \frac{x}{6}} \right]_{\pi/6}^{3\pi/2}$

$= -\frac{3}{2} \left\{ \frac{1}{\tan^4 \frac{3\pi/2}{6}} - \frac{1}{\tan^4 \left(\frac{\pi}{6}\right)} \right\}$

$= -\frac{3}{2} (1 - 9)$

$= 12$

(Answer)

3

$$\text{Let, } I_n = \int \sin^n x \, dx$$

$$= \int \sin^{n-1} x \sin x \, dx$$

$$= \sin^{n-1} x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \int \sin x \, dx \right\} dx$$

[Using the formula,

$$\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx]$$

$$= \sin^{n-1} x (-\cos x) - \int (n-1) \sin^{n-2} x \cos x (-\cos x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$\Rightarrow I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow I_n = \frac{1}{n} [-\sin^{n-1} x \cos x + (n-1) I_{n-2}]$$

$$\therefore \int \sin^n x dx = \frac{1}{n} [-\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx]$$

This is the reduction formula for $\int \sin^n x dx$.

Evaluating. $\int \sin^6 x \, dx = \frac{1}{6} \left[\sin^{6-1} x \cos x + (6-1) \int \sin^{6-2} x \, dx \right]$

$$= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \int \sin^4 x \, dx$$

$$= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \cdot \frac{1}{4} (-\sin^3 x) \cos x + \frac{5}{6} \cdot \frac{3}{2} \int \sin^2 x \, dx$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{128} \cdot \frac{1}{2} (-\sin x \cos x)$$

$$+ \frac{5}{28} \cdot \frac{1}{2} \int \sin^0 x \, dx$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x - \frac{5}{16} \sin x \cos x + \frac{5}{24} x + c$$

(Answer)

Height = 3 (the maximum value of $f(x)$ at $x=0$)

so the area under the curve,

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times 3 \times 3$$

$$= \frac{9}{2} \text{ square units}$$

4

Given that: $f(x) = 2 - |x|$, $[-1, 1]$

Let, $y = f(x)$

When, $x \geq 0$, $y = 2 - (+1) = 1$ $2 - x$; slope: $m = -1$

$x < 0$, $y = 2 + x$; slope $m = +1$

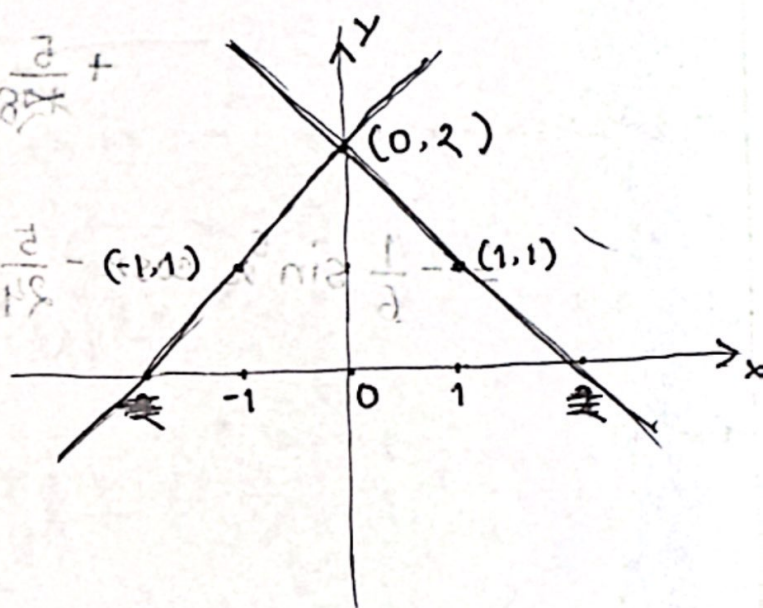
$x = -1$, $y = 2 + (-1) = 1$

$x = 1$, $y = 2 - 1 = 1$

Sketching the graph,

Here, we find the graph is an isosceles triangle.
Base = $1 + 1 = 2$

Height = 2 (the maximum value of $f(x)$ at $x = 0$)



So the area under the curve,

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times 2 \times 2$$

$$= 2 \text{ square units.}$$

5

Given that,

$$\int_1^e \frac{dy}{y\sqrt{1+(\ln y)^2}}$$

let, $u = \ln y$

$$du = \frac{1}{y} dy$$

we get,

$$\int \frac{du}{\sqrt{1+u^2}} = \int \frac{du}{\sqrt{u^2+1^2}} = \frac{1}{x} = \frac{1-x-x^2}{(1-x)^2 x}$$

$$= \ln \left[u + \sqrt{u^2+1^2} \right] = \ln \left[\ln y + \sqrt{(\ln y)^2+1^2} \right]$$

From the formula,

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln \left[x + \sqrt{x^2+a^2} \right]$$

Now,

$$\int_1^e \frac{dy}{y\sqrt{1+(\ln y)^2}} = \left[\ln \left[\ln y + \sqrt{(\ln y)^2+1^2} \right] \right]_1^e$$

$$= \ln \left[\ln e + \sqrt{(\ln e)^2+1^2} \right] - \ln \left[\ln 1 + \sqrt{(\ln 1)^2+1^2} \right]$$

$$= \ln \left[1 + \sqrt{2} \right] - 0$$

$$= \ln (1 + \sqrt{2})$$

(Answer)

6
Given. $\int \frac{2x^2 - x - 1}{x^3 - x^2} dx$

$$= \int \frac{2x^2 - x - 1}{x^2(x-1)} dx$$

Partial Fraction,

$$\frac{2x^2 - x - 1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} \quad \text{--- (1)}$$

$$\Rightarrow 2x^2 - x - 1 = A(x) + B + C(x-1)$$

$$\Rightarrow 2x^2 - x - 1 = (A+C)x^2 - (A-B)x - B$$

Equating the co-efficients,

$$A+C=2$$

$$A-B=1$$

$$B=1$$

$$A=2, B=1, C=0$$

$$\textcircled{1} \Rightarrow \frac{2x^2 - x - 1}{x^2(x-1)} = \frac{2}{x} + \frac{1}{x^2} + \frac{0}{x-1}$$

Now,

$$\int \frac{2x^2 - x - 1}{x^2(x-1)} dx = \int \left(\frac{2}{x} + \frac{1}{x^2} \right) dx$$

$$= 2 \ln|x| - \frac{1}{x} + c$$

(Answer)

$$\underline{\underline{\int_0^2 \frac{dx}{1-x^2}}}$$

In the point "1" the function will be discontinuous.

$$\int_0^2 \frac{dx}{1-x^2} = \int_0^1 \frac{dx}{1-x^2} + \int_1^2 \frac{dx}{1-x^2}$$

$$= \lim_{k \rightarrow 1^-} \int_0^k \frac{dx}{1-x^2} + \lim_{k \rightarrow 1^+} \int_k^2 \frac{dx}{1-x^2}$$

$$= \lim_{k \rightarrow 1^-} \left[\frac{1}{2} \ln \frac{1+x}{1-x} \right]_0^k + \lim_{k \rightarrow 1^+} \left[\frac{1}{2} \ln \frac{1+x}{1-x} \right]_k^2$$

$$= \lim_{k \rightarrow 1^-} \left(\frac{1}{2} \ln \frac{1+k}{1-k} - \frac{1}{2} \ln \frac{1+0}{1-0} \right) + \lim_{k \rightarrow 1^+} \left(\frac{1}{2} \ln \frac{1+2}{1-2} - \frac{1}{2} \ln \frac{1+k}{1-k} \right)$$

$$= \lim_{k \rightarrow 1^-} \left(\frac{1}{2} \ln \frac{1+k}{1-k} - 0 \right) + \lim_{k \rightarrow 1^+} \left(\frac{1}{2} \ln \frac{3}{-1} - \frac{1}{2} \ln \frac{1+k}{1-k} \right)$$

$$= \left(\frac{1}{2} \ln \frac{1+1}{1-1} \right) + \left(\frac{1}{2} \ln \frac{3}{-1} - \frac{1}{2} \ln \frac{1+1}{1-1} \right)$$

$$= \infty + \left(\frac{1}{2} \ln \frac{3}{-1} - \infty \right)$$

$$= \infty + (-\infty)$$

$$= \text{undefined}$$

$\therefore$ The integral is divergent.

8

$$\text{let, } x = e^{-t}$$

$$dx = -e^{-t} dt$$

$$\text{For } x=1, t=0$$

$$x=0, t=\infty$$

$$\begin{aligned} \int_0^1 x^m (\ln x)^n dx &= \int_{\infty}^0 (e^{-t})^m \cdot \{ \ln(e^{-t}) \}^n \cdot (-e^{-t}) dt \\ &= \int_0^{\infty} e^{-mt} (-1)^n t^n \cdot (-1) e^{-t} dt \\ &= (-1)^{n+1} \int_0^{\infty} e^{-(m+1)t} t^n dt \\ &= (-1)^{n+1} \left[\frac{t^n}{-(m+1)} - \frac{n}{m+1} \int_0^{\infty} e^{-(m+1)t} t^{n-1} dt \right] \\ &= (-1)^{n+1} \left[\frac{t^n}{-(m+1)} - \frac{n}{m+1} \left(\frac{t^{n-1}}{-(m+1)} - \frac{n-1}{m+1} \int_0^{\infty} e^{-(m+1)t} t^{n-2} dt \right) \right] \end{aligned}$$

From hint,

$$\int_0^{\infty} e^{-kx} x^{n-1} dx = \frac{\Gamma(n)}{k^n}$$

$$\int_0^1 x^m (\ln x)^n dx = (-1)^n \times (-1)^n \times (-1) \times \frac{\Gamma(n+1)}{(m+1)^{n+1}}$$

$$= \frac{(-1)^n \Gamma(n+1)}{(m+1)^{n+1}} = \left[\begin{matrix} k=m+1 \\ n=n+1 \end{matrix} \right]$$

$$\frac{\frac{1}{2} \cdot \frac{1}{2}}{\left(\frac{1}{2} + \frac{1}{2}\right)^2} = \frac{1}{2} \quad (\text{showed})$$

Given, $\int_0^1 x^6 (1-x^2)^{1/2} dx$

let, $x^2 = t \Rightarrow x = t^{1/2}$ For, $x=1, t=1$

$\therefore dt = 2x \cdot dx$ $x=0, t=0$

$\Rightarrow dx = \frac{dt}{2x}$

$$\int_0^1 x^6 (1-x^2)^{1/2} dx = \int_0^1 (t^{1/2})^6 (1-t)^{1/2} \frac{dt}{2t^{1/2}}$$

$$= \int_0^1 t^3 (1-t)^{1/2} \cdot \frac{1}{2} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^1 t^{5/2} (1-t)^{1/2} dt = \frac{1}{2} \int_0^1 t^{3/2-1} (1-t)^{3/2-1} dt$$

— (1)

Euler's law,

$$\beta(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt$$

$$\Rightarrow \frac{1}{2} \beta\left(\frac{7}{2}, \frac{3}{2}\right) = \frac{1}{2} \int_0^1 t^{\frac{7}{2}-1} (1-t)^{\frac{3}{2}-1} dt \quad [\text{comparing with (1)}]$$

We know,

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{7}{2}\right) \cdot \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{7}{2} + \frac{3}{2}\right)}$$

$$= \frac{1}{2} \times \frac{\frac{\sqrt{\pi}}{2} \times \frac{15\sqrt{\pi}}{8}}{24}$$

$$= \frac{1}{2} \times \frac{\frac{\sqrt{\pi}}{2} \times \frac{15\sqrt{\pi}}{8}}{24} \quad [\Gamma 5 = 4! = 24]$$

$$= \frac{1}{2} \times \frac{15\pi}{16} \times \frac{1}{24}$$

$$= \frac{5\pi}{256} \quad (\text{Answer})$$